

**Part 1** If we're trying to predict the results of the Clinton vs. Trump presidential race, what is the population of interest?

The population of interest would be the citizens of America who are eligible to vote.



**Part 2** What is the sampling frame?

The sampling frame would be the people who have valid cell phone numbers that were drawn from the random digit dialing method.



### 0.0.1 Question 5

Why can't we assess the impact of the other two biases (voters changing preference and voters hiding their preference)?

Note: You might find it easier to complete this question after you've completed the rest of the homework including the simulation study.

We can't assess it because it's confidential information. People might lie in the survey, or be reluctant to give out their political opinion.

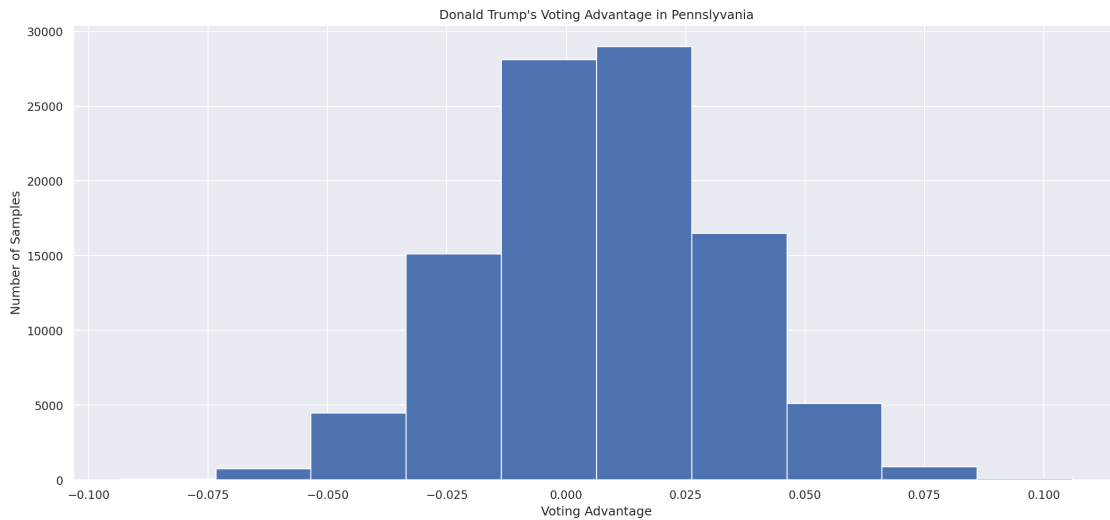


**Part 4** Make a histogram of the sampling distribution of Trump's proportion advantage in Pennsylvania. Make sure to give your plot a title and add labels where appropriate. Hint: You should use the `plt.hist` function in your code.

Make sure to include a title as well as axis labels. You can do this using `plt.title`, `plt.xlabel`, and `plt.ylabel`.

```
In [38]: plt.hist(simulations)
         plt.title("Donald Trump's Voting Advantage in Pennsylvania")
         plt.xlabel("Voting Advantage")
         plt.ylabel("Number of Samples")
```

```
Out[38]: Text(0, 0.5, 'Number of Samples')
```





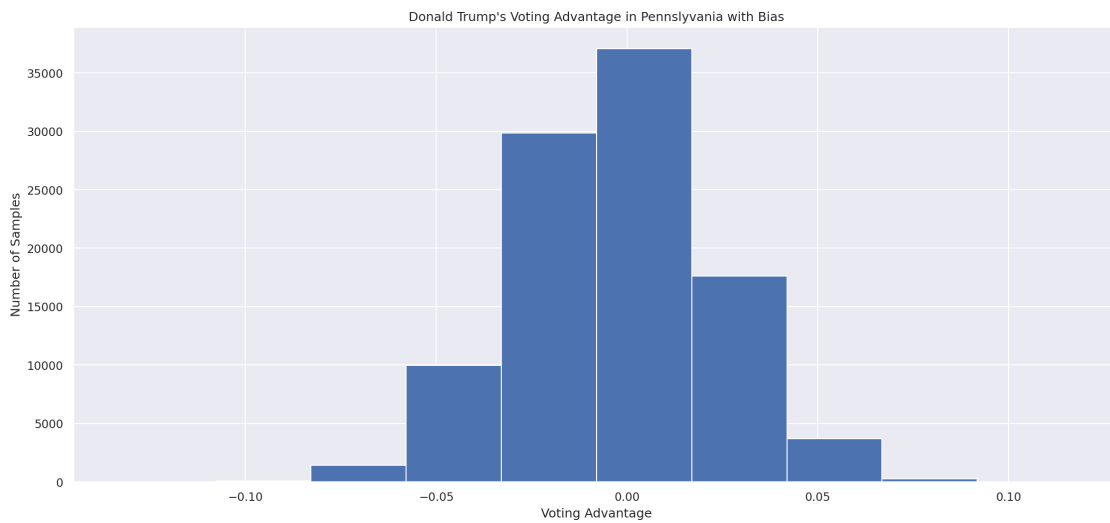


**Part 2** Make a histogram of the new sampling distribution of Trump's proportion advantage now using these biased samples. That is, your histogram should be the same as in Q6.4, but now using the biased samples.

Make sure to give your plot a title and add labels where appropriate.

```
In [45]: plt.hist(biased_simulations)
plt.title("Donald Trump's Voting Advantage in Pennsylvania with Bias")
plt.xlabel("Voting Advantage")
plt.ylabel("Number of Samples")
```

```
Out[45]: Text(0, 0.5, 'Number of Samples')
```





**Part 3** Compare the histogram you created in Q7.2 to that in Q6.4.

It is nearly the same as before, where the biased simulations are shifted to the left more.



Write your answer in the cell below.

Increasing the sample size actually amplifies the numbers to the point that more sample size increases the chances that Trump will win, and decreases the chances in a biased situation. More so, increasing the sample size might've decreased the error rate. If done right, a large sample size might converge to the more truthful prediction.



### 0.0.2 Question 9

According to FiveThirtyEight: “... Polls of the November 2016 presidential election were about as accurate as polls of presidential elections have been on average since 1972.”

When the margin of victory may be relatively small as it was in 2016, why don't polling agencies simply gather significantly larger samples to bring this error close to zero?

Gathering larger samples takes more resources and effort, which might be hard to do.

